SMTP, POP3, IMAP

Application layer

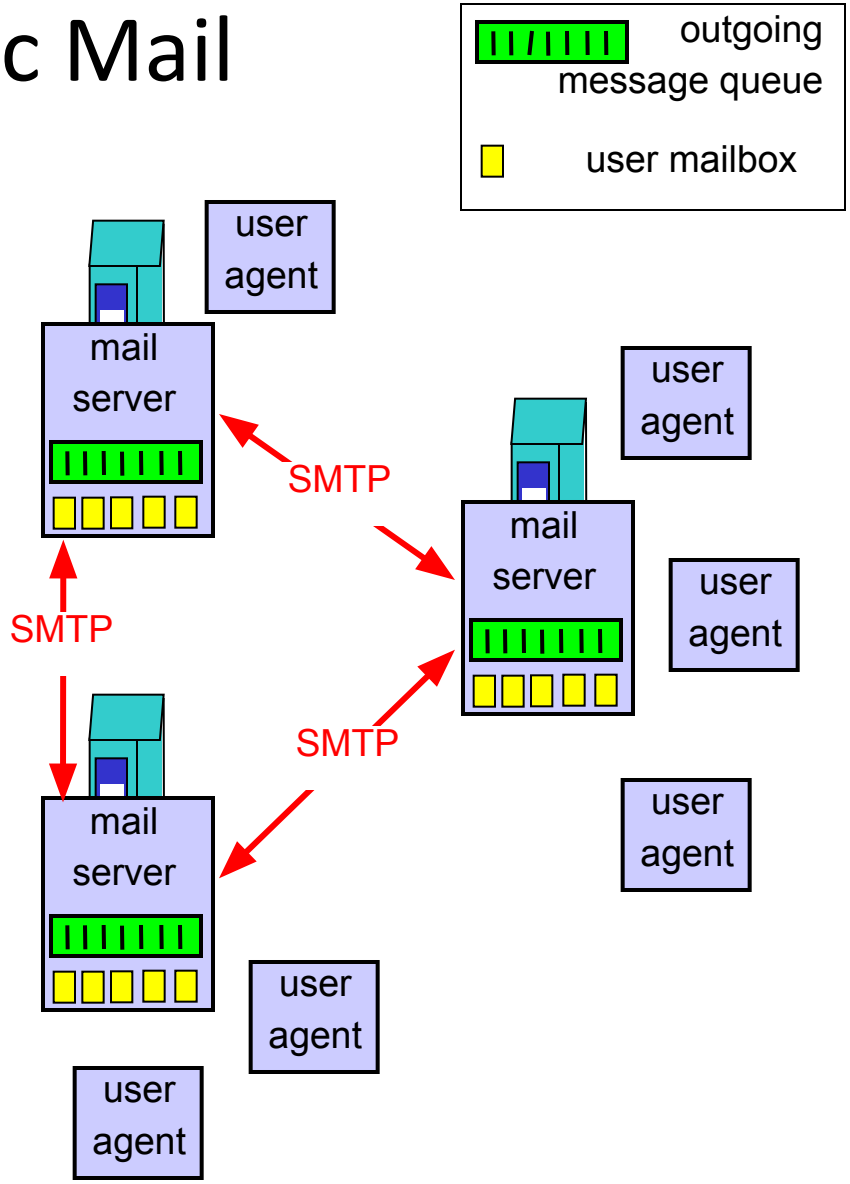
- 2.1 Principles of network applications
- 2.2 Web and HTTP
- 2.3 FTP
- Electronic Mail
  - SMTP, POP3, IMAP
- DNS
- 2.6 P2P applications
- 2.7 Socket programming with TCP
- 2.8 Socket programming with UDP

Electronic Mail

- Three major components:
  - user agents
  - mail servers
  - simple mail transfer protocol: SMTP

User Agent

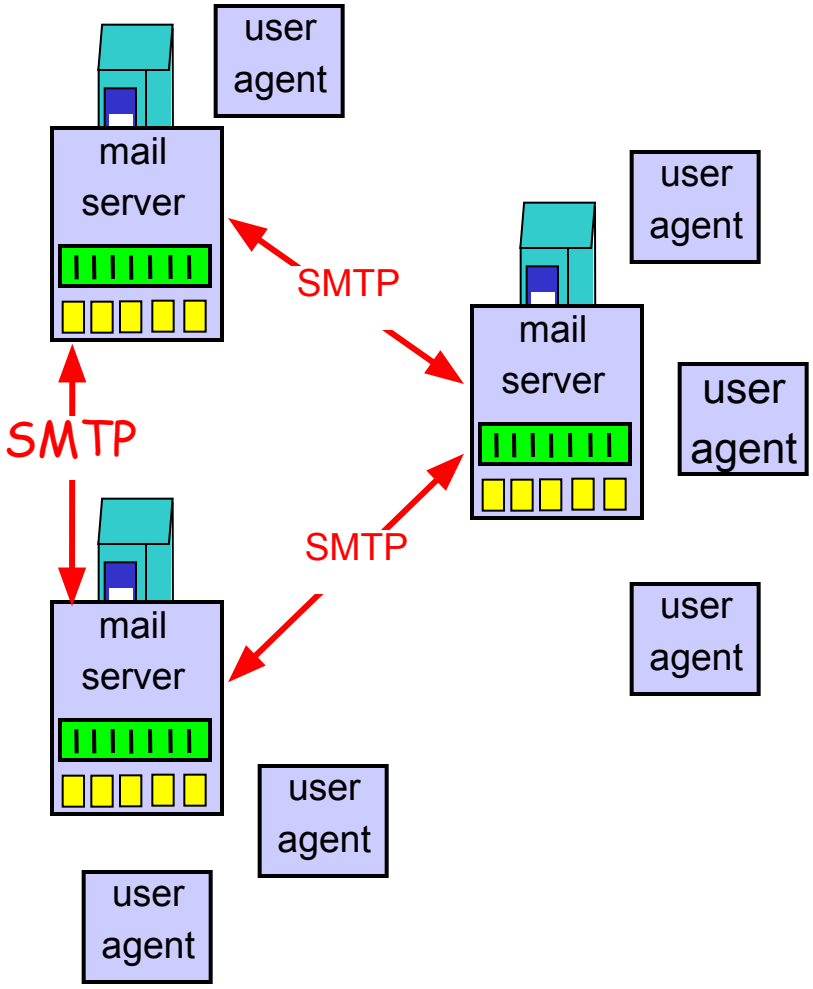
- a.k.a. “mail reader”
- composing, editing, reading mail messages
- e.g., Eudora, Outlook, elm, Mozilla Thunderbird, browser based.
- outgoing, incoming messages stored on server



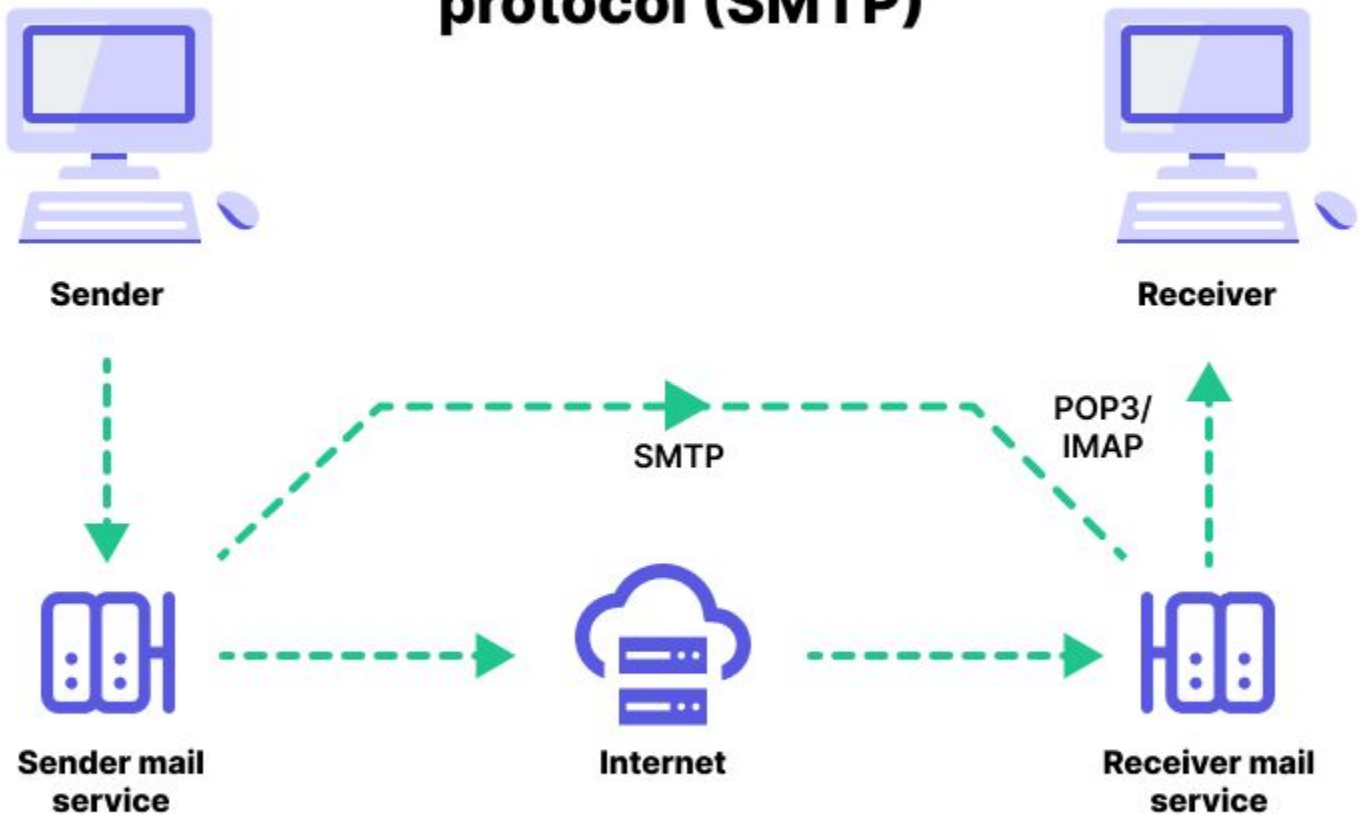
Electronic Mail: mail servers

- Mail Servers

- mailbox contains incoming messages for user
- message queue of outgoing (to be sent) mail messages
- SMTP protocol between mail servers to send email messages
  - client: sending mail server
  - "server": receiving mail server



Simple mail transfer protocol (SMTP)



Simple Mail Transfer Protocol (SMTP)

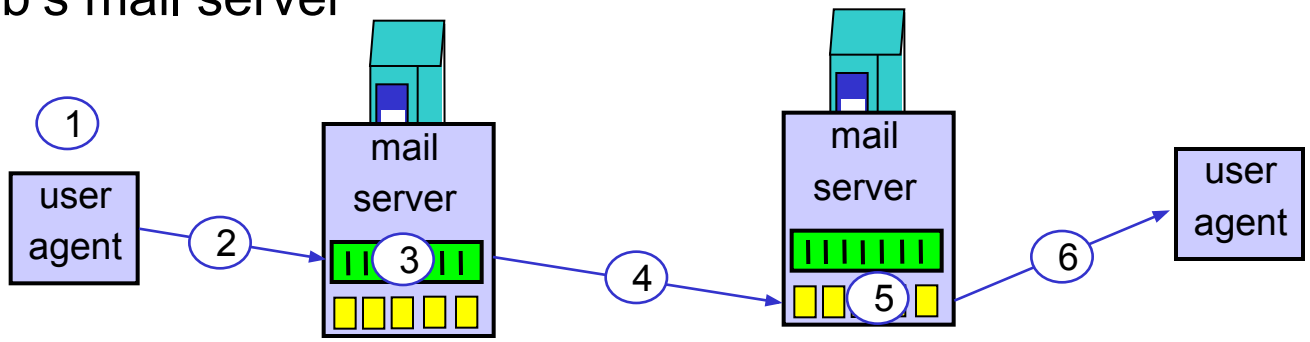
- **Definition:**
  SMTP is the Internet standard communication protocol for **sending and relaying email messages** across networks — like a **postal service for email**.
- **Function:**
  - Handles **outbound email sending** from a client to a mail server.
  - Transfers emails **between mail servers** until delivery to the recipient.
- **How It Works:**
  - The email client establishes a **TCP connection** to an SMTP server.
  - The SMTP server **relays the message** to the next mail server.
  - The process continues until the **recipient's mail server** receives it.
- **Key Components:**
  - **SMTP Server:** Acts as the "post office" for sending mail.
  - **Mail Transfer Agent (MTA):** Relays messages between servers.
  - **Port Numbers:** Commonly uses **25**, **465**, or **587**.
  - **Email Clients:** Gmail, Outlook, Apple Mail use SMTP for sending.

Electronic Mail: SMTP [RFC 2821]

- uses TCP to reliably transfer email message from client to server, port 25
- direct transfer: sending server to receiving server
- three phases of transfer
  - handshaking (greeting)
  - transfer of messages
  - closure
- command/response interaction
  - commands: ASCII text
  - response: status code and phrase
- messages must be in 7-bit ASCII

Scenario: Alice sends message to Bob

- 1) Alice uses UA to compose message and "to" bob@someschool.edu
- 2) Alice's UA sends message to her mail server; message placed in message queue
- 3) Client side of SMTP opens TCP connection with Bob's mail server
- 4) SMTP client sends Alice's message over the TCP connection
- 5) Bob's mail server places the message in Bob's mailbox
- 6) Bob invokes his user agent to read message



Sample SMTP interaction

```
S: 220 somecollege.edu
C: HELO anothercollege.fr
S: 250  Hello anothercollege.fr, pleased to meet you
C: MAIL FROM: <alice@ anothercollege.fr>
S: 250 alice@ anothercollege.fr... Sender ok
C: RCPT TO: <bob@ somecollege.edu>
S: 250 bob@somecollege.edu ... Recipient ok
C: DATA
S: 354 Enter mail, end with "." on a line by itself
C: Do you want to share feedback?
C: How about additional points?
C: .
S: 250 Message accepted for delivery
C: QUIT
S: 221 somecollege.edu closing connection
```

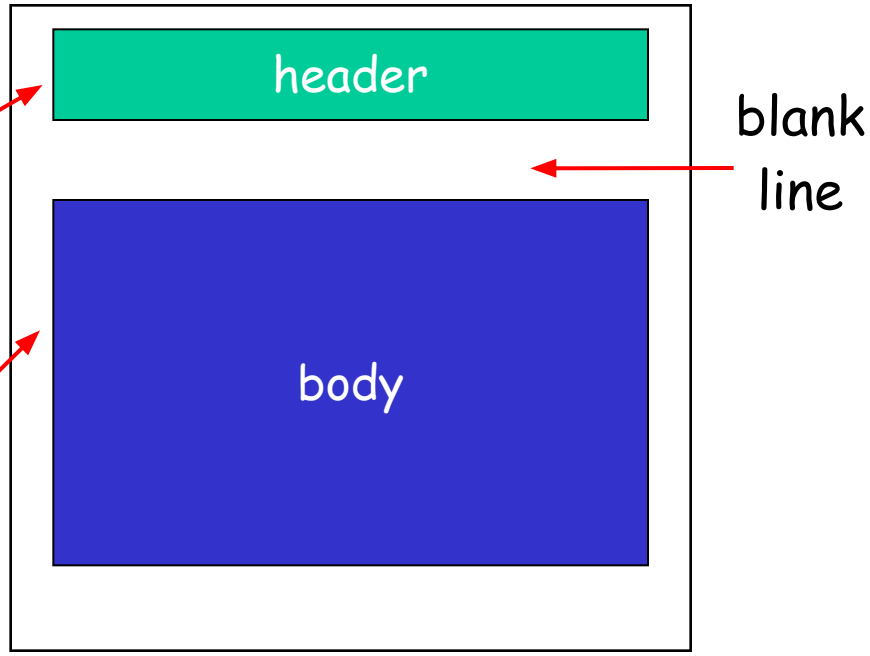
Mail message format

- SMTP: protocol for exchanging email msgs
- RFC 822: standard for text message format:

- header lines, e.g.,
  - To:
  - From:
  - Subject:

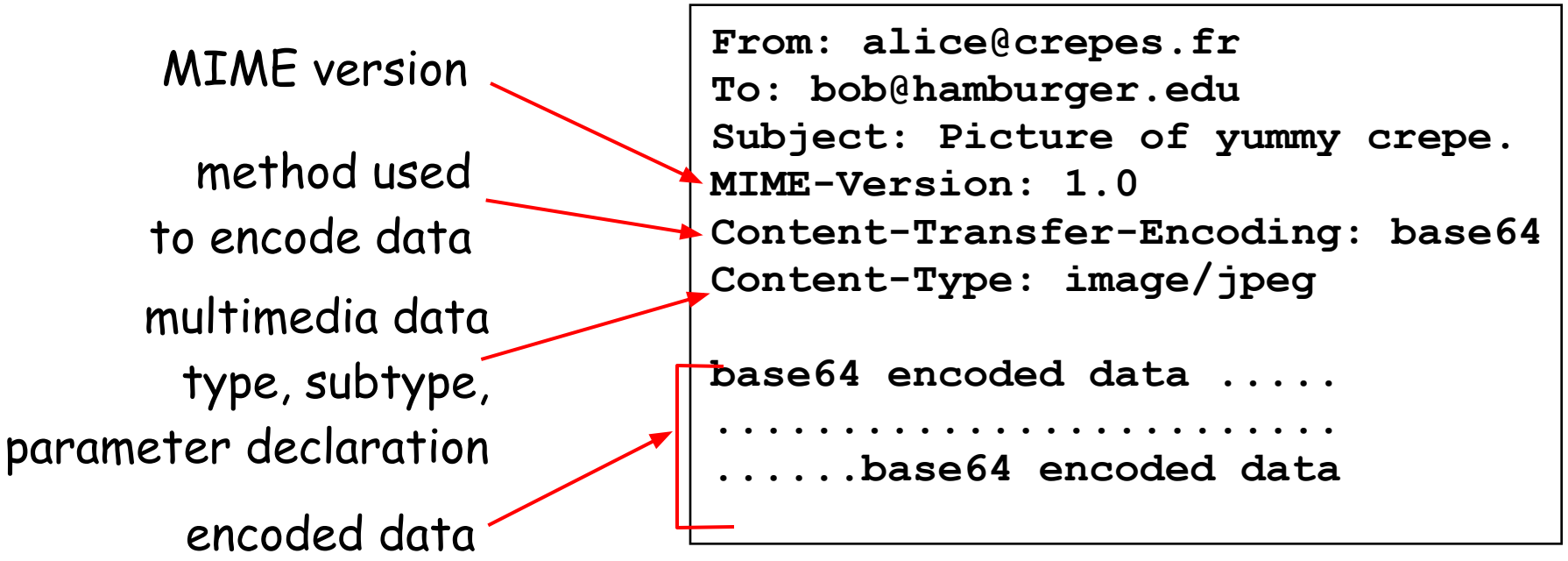
different from SMTP commands!

- body
  - the “message”, ASCII characters only

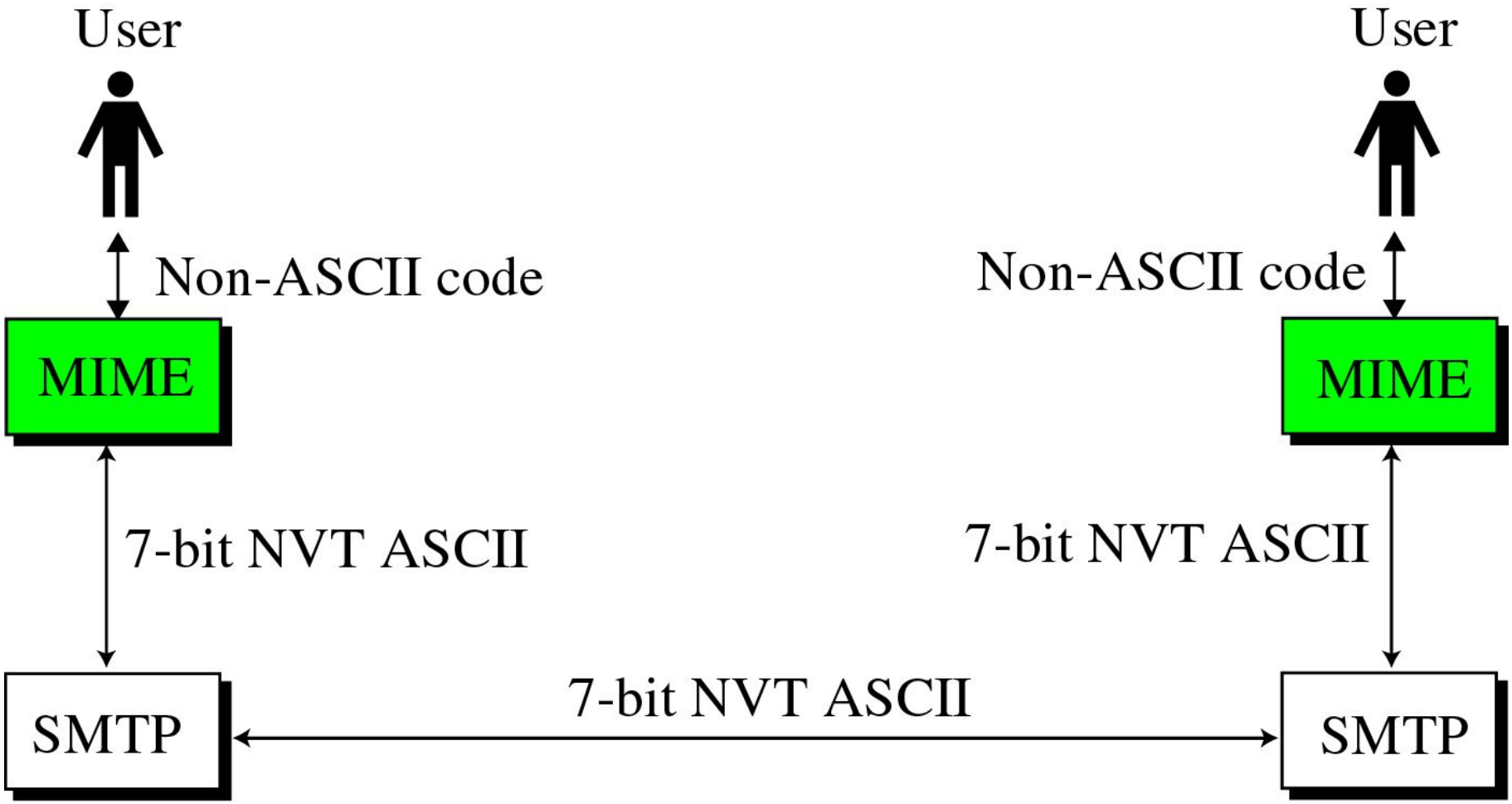


Message format: multimedia extensions

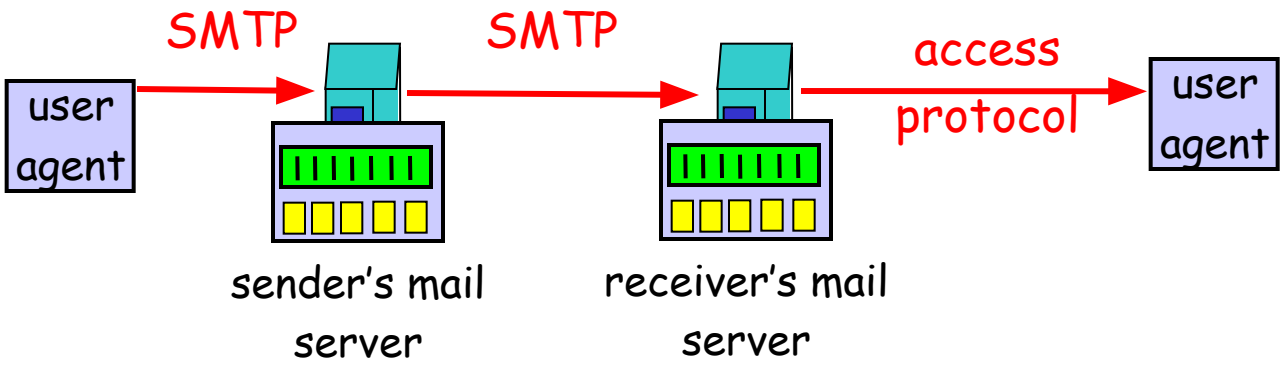
- MIME: multimedia mail extension, RFC 2045, 2056
- additional lines in msg header declare MIME content type



MIME

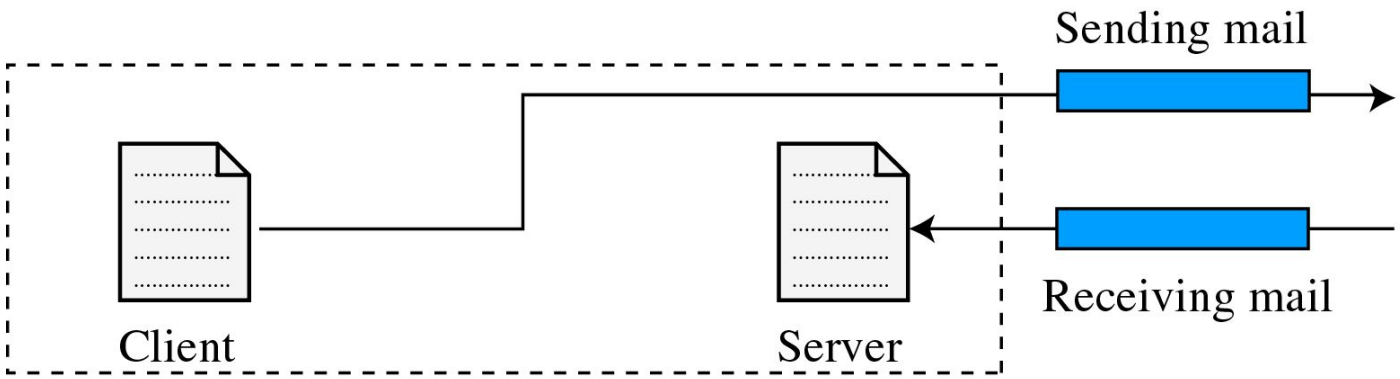

User
User
Non-ASCII code
Non-ASCII code
MIME
MIME
7-bit NVT ASCII
7-bit NVT ASCII
SMTP
7-bit NVT ASCII
SMTP

Mail access protocols

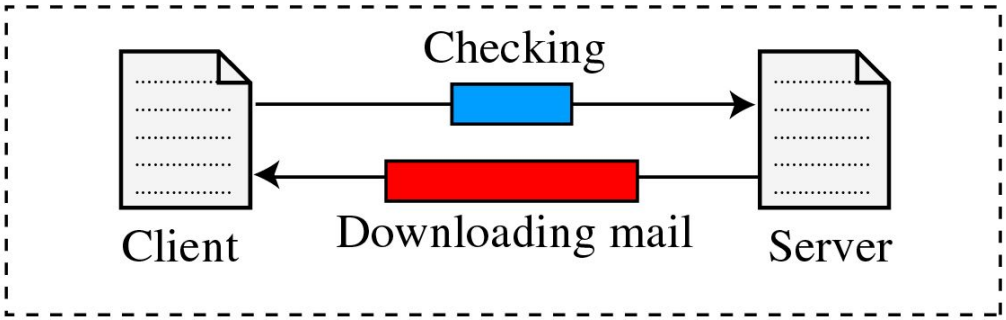


- SMTP: delivery/storage to receiver's server
- Mail access protocol: retrieval from server
  - POP: Post Office Protocol [RFC 1939]
    - authorization (agent <-->server) and download
  - IMAP: Internet Mail Access Protocol [RFC 1730]
    - more features (more complex)
    - manipulation of stored msgs on server
  - HTTP: gmail, Hotmail, Yahoo! Mail, etc.

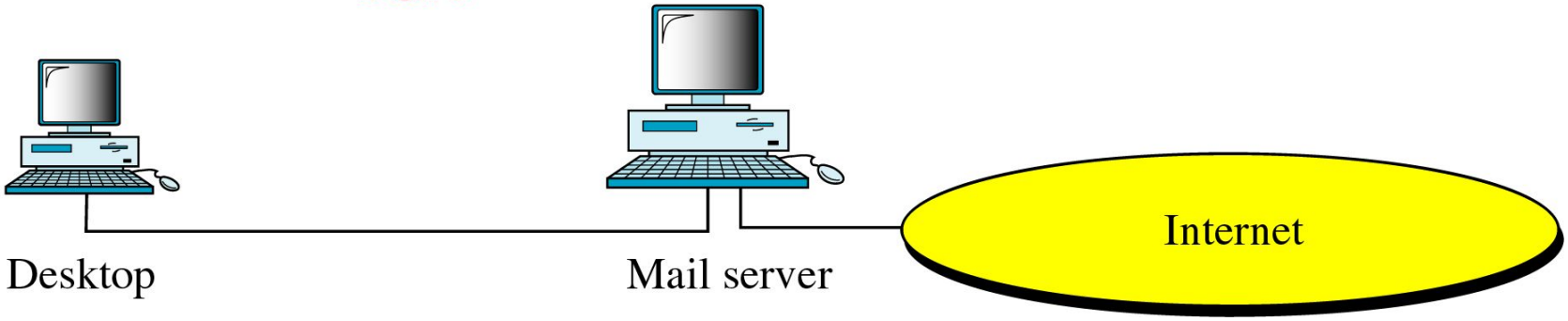
POP3 and SMTP



SMTP



POP3



POP3 protocol

- authorization phase
  - client commands:
    - **user:** declare username
    - **pass:** password
  - server responses
    - **+OK**
    - **-ERR**
- transaction phase, client:
  - **list:** list message numbers
  - **retr:** retrieve message by number
  - **dele:** delete
  - **quit**

```
S: +OK POP3 server ready
C: user bob
S: +OK
C: pass hungry
S: +OK user successfully logged on

C: list
S: 1 498
S: 2 912
S: .
C: retr 1
S: <message 1 contents>
S: .
C: dele 1
C: retr 2
S: <message 1 contents>
S: .
C: dele 2
C: quit
S: +OK POP3 server signing off
```

POP3 (more) and IMAP

- More about POP3
  - Previous example uses "download and delete" mode.
  - Bob cannot re-read e-mail if he changes client
  - "Download-and-keep": copies of messages on different clients
  - POP3 is stateless across sessions

- IMAP
  - Keep all messages in one place: the server
  - Allows user to organize messages in folders
  - IMAP keeps user state across sessions:
    - names of folders and mappings between message IDs and folder name

SMTP, FTP and HTTP

- Connections:
  - All use TCP
  - SMTP uses persistent connections
  - FTP persistent control channel, non-persistent data channel per file.
  - HTTP uses both persistent and non-persistent channels
- Data Flow
  - FTP and HTTP: pull (user requests files)
  - SMTP: push (email sent to receiver without request, result can be spam!)

- All have ASCII command/response interaction, status codes
- HTTP: each object encapsulated in its own response msg
- FTP: one data channel per file
- SMTP: multiple objects sent in multipart msg

Comparison between POP3 and IMAP

- POP3 (Post Office Protocol) and IMAP are both email protocols that allow users to access their email, but they differ in several ways:
- **Number of devices**: POP3 is designed for use on a single device, while IMAP is designed for use on multiple devices.
- **Email storage**: POP3 downloads emails to a local device and deletes them from the server, while IMAP stores emails on the server and syncs them across devices.
- **Email organization**: IMAP allows users to organize emails on the server, while POP3 does not.
- **Email updates**: IMAP allows users to create and update emails on the server, while POP3 does not.
- **Email access**: IMAP allows users to access emails in real-time across multiple devices, while POP3 only allows access to emails on a single device.
- **Email synchronization**: IMAP automatically syncs changes made to emails across all devices, while POP3 does not.
- **Email security**: IMAP uses TLS or SSL encryption, while POP3 is unencrypted.
- IMAP is generally recommended over POP3 because it's more adaptable and sophisticated, and it's better for managing emails across multiple devices. However, POP3 may be a better option if you're only using one device, have a slow internet connection, or want to access emails offline